

A finite-measure subcase of the non-radial Hilbert-matrix norm problem

arXiv automatic math run `fa_banach_001`

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Source and question

The source paper is Bellavita–Stylogiannis, *Hilbert matrix operator Acting between conformally invariant spaces*, arXiv:2408.01060 [1]. Its final section records the following question.

8. OPEN QUESTIONS

We note that for every radial measure μ that satisfies (2.1),

$$\|\mathcal{H}\|_{H^\infty \rightarrow M(\mathcal{D}_\mu)} = 1 + \left(\int_{\mathbb{D}} \frac{4}{|1 - z^2|^2} U_\mu(z) dA(z) \right)^{1/2}.$$

However we are not able to compute the exact value of $\|\mathcal{H}\|_{H^\infty \rightarrow M(\mathcal{D}_\mu)}$ in general.

Question 8.1. What is the exact norm of $\mathcal{H}$ from H^∞ to $M(\mathcal{D}_\mu)$ for non-radial μ ?

In the notation of the source, for a positive Borel measure μ on $\mathbb{D}$,

$$U_\mu(z) = \int_{\mathbb{D}} \log \left| \frac{1 - \bar{w}z}{z - w} \right|^2 d\mu(w)$$

and

$$\|g\|_{M(\mathcal{D}_\mu)} = |g(0)| + \sup_{\phi \in \text{Aut}(\mathbb{D})} \left(\int_{\mathbb{D}} |g'(\phi(z))|^2 U_\mu(\phi(z)) dA(z) \right)^{1/2}.$$

The source proves the exact formula for radial μ . The question asks for the exact norm when μ is non-radial.

Result

Theorem 1. *Let μ be a finite nonzero positive Borel measure on $\mathbb{D}$. Then*

$$\|\mathcal{H}\|_{H^\infty \rightarrow M(\mathcal{D}_\mu)} = 1 + \frac{\pi}{\sqrt{2}} \sqrt{\mu(\mathbb{D})}.$$

In particular, Question 8.1 has this exact answer for every finite non-radial measure.

Proof intuition

For finite μ , the $M(\mathcal{D}_\mu)$ -seminorm can be written as an average of the usual local BMOA Green seminorms, after moving the point of evaluation by automorphisms. Degenerating automorphisms toward the boundary send every fixed point of $\mathbb{D}$ to the same boundary point. Thus, for the logarithm $L(z) = \log(1/(1-z))$, all local seminorms approach the extremal BMOA value $\pi/\sqrt{2}$ simultaneously, and finiteness of μ allows dominated convergence through the integral over w .

The upper bound for $\mathcal{H}$ is exactly the same estimate used in the source: $(1-z)(\mathcal{H}f)'(z)$ is bounded by $\|f\|_\infty$. The lower bound is obtained by testing $f \equiv 1$. The function $\mathcal{H}(1)$ has the same boundary local BMOA limit as L , so the upper bound is sharp.

Proof

Write

$$G(z, w) = \log \left| \frac{1 - \bar{w}z}{z - w} \right|^2.$$

For an analytic function g , set

$$\mathcal{D}_a(g) = \int_{\mathbb{D}} |g'(z)|^2 G(z, a) dA(z), \quad a \in \mathbb{D}.$$

The Green kernel is invariant under automorphisms:

$$G(\phi(z), w) = G(z, \phi^{-1}(w)), \quad \phi \in \text{Aut}(\mathbb{D}).$$

Therefore, by Fubini,

$$\int_{\mathbb{D}} |g'(z)|^2 U_\mu(\phi(z)) dA(z) = \int_{\mathbb{D}} \mathcal{D}_{\phi^{-1}(w)}(g) d\mu(w). \quad (1)$$

Let

$$L(z) = \log \frac{1}{1-z}.$$

The classical computation used in the source paper's proof of the $H^\infty \rightarrow \text{BMOA}$ norm gives

$$\sup_{a \in \mathbb{D}} \mathcal{D}_a(L)^{1/2} = \frac{\pi}{\sqrt{2}}, \quad \mathcal{D}_{\sigma_r(w)}(L)^{1/2} \rightarrow \frac{\pi}{\sqrt{2}}$$

for every fixed $w \in \mathbb{D}$, as $r \uparrow 1$, where $\sigma_r(z) = (r-z)/(1-rz)$. Applying (1) and dominated convergence gives

$$\sup_{\phi \in \text{Aut}(\mathbb{D})} \left(\int_{\mathbb{D}} |L'(z)|^2 U_\mu(\phi(z)) dA(z) \right)^{1/2} = \frac{\pi}{\sqrt{2}} \sqrt{\mu(\mathbb{D})}. \quad (2)$$

For $f \in H^\infty$, the source proves the identity

$$(1-z)(\mathcal{H}f)'(z) = b_f(z), \quad b_f \in H^\infty, \quad \|b_f\|_\infty \leq \|f\|_\infty.$$

Since $\mathcal{H}f(0) = \int_0^1 f(t) dt$, (2) gives

$$\|\mathcal{H}f\|_{M(\mathcal{D}_\mu)} \leq \left(1 + \frac{\pi}{\sqrt{2}} \sqrt{\mu(\mathbb{D})} \right) \|f\|_\infty.$$

Thus the displayed quantity is an upper bound for the operator norm.

It remains to see that it is sharp. Put

$$K(z) = \mathcal{H}(1)(z) = \frac{1}{z} \log \frac{1}{1-z},$$

with the removable value $K(0) = 1$. The same boundary computation used in Bellavita–Stylogiannis for the lower estimate gives

$$\sup_{a \in \mathbb{D}} \mathcal{D}_a(K)^{1/2} = \frac{\pi}{\sqrt{2}}, \quad \mathcal{D}_{\sigma_r(w)}(K)^{1/2} \longrightarrow \frac{\pi}{\sqrt{2}}$$

for every fixed $w \in \mathbb{D}$ as $r \uparrow 1$. One way to see the limit is to use the source decomposition

$$zK'(z) = -\frac{1}{z} \log \frac{1}{1-z} + \frac{1}{1-z};$$

the first term is negligible in the degenerating Green seminorm, while the second has the same boundary limit as L' . Again (1) and dominated convergence imply

$$\sup_{\phi \in \text{Aut}(\mathbb{D})} \left(\int_{\mathbb{D}} |K'(\phi(z))|^2 U_{\mu}(\phi(z)) dA(z) \right)^{1/2} = \frac{\pi}{\sqrt{2}} \sqrt{\mu(\mathbb{D})}.$$

Since $K(0) = 1$ and $\|1\|_{H^\infty} = 1$,

$$\|\mathcal{H}\|_{H^\infty \rightarrow M(\mathcal{D}_\mu)} \geq \|\mathcal{H}(1)\|_{M(\mathcal{D}_\mu)} = 1 + \frac{\pi}{\sqrt{2}} \sqrt{\mu(\mathbb{D})}.$$

Together with the upper bound, this proves the theorem.

Verification notes and limitations

The proof uses only the source paper’s Hilbert-matrix derivative identity and its standard BMOA boundary computation for L and $\mathcal{H}(1)$, plus the Green-kernel invariance formula. The finite-measure hypothesis is essential for the dominated-convergence step as written. The result does not settle the remaining infinite non-radial measures in Question 8.1.

The local run indexes had no exact prior packet for arXiv:2408.01060. A bounded web search on 2026-06-18 for the arXiv id, the paper title, and phrases including “non-radial”, “Hilbert matrix”, and $M(\mathcal{D}_\mu)$ found the source paper and nearby Hilbert-matrix papers/surveys, but no later exact answer to Question 8.1.

Human-review recommendation

Review as a likely valid partial answer. The main point to check is the asserted boundary-local BMOA limit for $K = \mathcal{H}(1)$, especially that the negligible logarithmic term remains dominated uniformly enough to justify the finite-measure dominated-convergence passage.

References

- [1] C. Bellavita and G. Stylogiannis, *Hilbert matrix operator Acting between conformally invariant spaces*, arXiv:2408.01060, 2024.